Challenge Brief

Template for challenge partners · Since AI Hackathon 2026 · 6–8 Nov 2026, Turku

This template helps you define a clear, buildable challenge for the hackathon. Fill in each section below — the more concrete the problem and the evaluation criteria, the better the prototypes you receive. You can return it to us as is, and we are happy to help shape it into a ready event challenge.

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Challenge partner (company)

Meyer Turku Oy

01 Challenge Title

A short, memorable name for the challenge. Example: "Automated support ticket routing."

Automated class and regulatory compliance checking

02 Problem Statement

What business or technical problem should be solved? Describe the context, why it matters, and the main pain points.

Cruise ships, like the ones we build at Meyer Turku are subject to many rules and regulations, including:

- SOLAS *(International Convention for the Safety of Life at Sea)*
- MARPOL *(International Convention for the Prevention of Pollution from Ships)*
- COLREG *(International Regulations for Preventing Collisions at Sea)*
- Load Line Convention
- Tonnage Convention
- Classification Society Rules, e.g. DNV
- PVAG *(Passenger Vessel Accessibility Guidelines)*
- ADA (Americans with Disabilities Act)

In ship projects, it is important to establish compliance with these rules and regulations early in the ship design phase. Traditionally, this is manual work that requires a high level of diligence, experience and a lot of time to get right – time that could be spent on engineering challenges.

We are looking for a solution that assists our engineers to check designs against class rules (e.g. DNV), SOLAS and regulations early and inform them of any issues found in the design.

03 Guiding Question

One question that focuses the challenge. Example: "How might we use AI to predict equipment failures before they happen?"

How could designs be checked against class rules (e.g. DNV), SOLAS and regulations early, before conflicts propagate downstream?

04 Recommended AI Approaches *Optional*

Any relevant technologies, models or methods — for example NLP for text analysis, classification, knowledge graphs, agents, or RAG.

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05 Dataset or API Provided *If any*

Describe any data or resources you provide: format, a link if ready, and sample size. If none, state what builders should expect.

> At Since AI, challenges can be built with synthetic, anonymized or public data — no personal data is required.

While we cannot provide any actual design files from our operations, participants can visit the following links for examples of layouts and technical documentation from actual ships:

- (3D model & GA) https://ceoas.oregonstate.edu/regional-class-research-vessel-ship-design
- https://www.unols.org/regional-class-research-vessel-rcrv-update-project
- (Only Study) https://repository.library.noaa.gov/view/noaa/30507/noaa_30507_DS1.pdf
- (GA) https://dana.dtu.dk/-/media/subsites/dana/23007-01-101-0110-01-r2-14-general-arrangement.pdf
- (GA) https://www.atlanticship.dk/wp-content/uploads/2020/03/9281-ga-plan-2017117-01-model-anynomized.pdf
- (GA) https://www.hi.no/resources/04650-Rev.H-General-Arrangement-GO-SARS.pdf
- http://www.perseus-net.eu/assets/media/PDF/deliverables/6790.4_Final.pdf

Rules and regulations are not always published in full, but summaries are often available:

Regulation / Standard	Available Public	Notes
SOLAS	Partially	IMO publishes summaries and guidance publicly.
MARPOL	Partially	Same as SOLAS. Annex summaries are public
COLREG	Yes	The convention text is widely available from government and maritime authority websites.
Load Line Convention	Partially	Summaries are public, complete official text generally through IMO publications.
Ballast Water Management Convention	Partially	Core requirements available publicly; official consolidated text through IMO.
Classification Society Rules e.g. DNV	Yes	Through DNV's Veracity portal

Accessibility Guidelines (PVAG)	Yes	U.S. Passenger Vessel Accessibility Guidelines are freely available from U.S. government sources.
Accessibility Guidelines (ADA)	Yes	Freely available from U.S. government sources.

06 Evaluation Criteria

How should solutions be judged? What matters most — for example accuracy, scalability, usability, or business impact?

Accuracy is very important to us, specifically in the context of rules & regulations; good usability is a plus. It is important that you can demonstrate that your solution can properly interpret design documents (written text / technical drawings / 3D models) and highlight why something is not compliant with a specific aspect of a standard. A solid foundation is more crucial than a polished product.

07 Submission Format

What deliverables do you expect? For example a demo video, pitch slides, a code repository, or documentation.

We like to see a short demo, code repository and documentation that covers deployment

08 Point of Contact

Name, email and preferred contact method for questions during the hackathon.

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